



NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
BSc (Hons) Computing|ITB|EHN|Year-1
DIPLOMA IN SOFTWARE ENGINEERING|-23.1
DATABASE MANAGEMENT SYSTEMS
2023-09-27

This paper contains **06** questions. Read instructions carefully.

Answer All questions.

Time: **THREE** hours.

This paper contains only 06 questions. Answer all questions. Other than this exam, student's knowledge is evaluated through practical course work.

The student's practical knowledge is evaluated during the Practical Course Work. For the Practical course work students must get above 40 marks to pass this module.

Question One (Underline the correct answer in this paper). If you don't underline the answers in this paper you won't get any marks for question one.
(Each carry 02 Marks. 2 * 10 = 20 Marks)

Question 1 to 5 based on below trigger

CREATE TRIGGER Trg_EMP_audit

ON Employee

AFTER INSERT, DELETE

AS BEGIN

INSERT INTO Employee_audits (Eno, Ename, Address, Position, Salary, Dno, Updated_at, Operation)

SELECT Eno, Ename, Address, Position, Salary, Dno, GETDATE(), 'INS'

FROM INSERTED

UNION ALL

SELECT Eno, Ename, Address, Position, Salary, Dno, GETDATE(), 'DEL'

FROM DELETED

END;

1. What is a trigger?
 - a) It is same as subquery where sa admin user can execute
 - b) Special stored procedures that are executed manually
 - c) Special stored procedures that are executed automatically
 - d) It is same as stored procedures where sa admin user can execute
2. Above trigger will execute
 - a) After Insert, Delete and Select on Employee table
 - b) After Insert or Delete on Insert table
 - c) After Insert or Delete on Employee_audits table
 - d) After Insert or Delete on Employee table
3. What are INSERTED and DELETED tables in above trigger?
 - a) Tables created by sa admin user that can capture the deleted data
 - b) Tables created by sa admin user that can capture the modify data
 - c) Virtual tables that can capture the modify data
 - d) Help Employee table to update data in Employee_audits table
4. What is the use of GETDATE() function in above trigger?
 - a) To enter system data and time by sa admin user
 - b) To enter system time automatically
 - c) To get the system data and time automatically
 - d) To enter system year automatically
5. What is the use of UNION ALL in above trigger?
 - a) Take intersection of INSERTED and DELETED tables
 - b) Combine the results set of INSERT and Employee_audits tables
 - c) Combine the results set of INSERTED and DELETED tables
 - d) Take intersection of INSERTED and Employee_audits tables

Question 6 to 10 based on below table

```
CREATE TABLE Employee_Audits (Change_id INT IDENTITY PRIMARY KEY, Eno  
varchar(10), Ename varchar(10), Address varchar(10), Position varchar, Salary int, Dno  
varchar(10), Updated_at DATETIME, Operation CHAR(3), CHECK(operation = 'INS'  
OR operation='DEL'));
```

6. What is the use of IDENTITY key word?

- e) Use to confirm that change_id is the primary key
- f) Use to automatically increment the primary key from 1 to 9 only
- g) Use to manually enter primary key from 1 to 1000
- h) Use to automatically increment the primary key

7. Can you chose Ename column as the primary key?

- a) Yes, because Ename cannot be blank
- b) Yes, because Ename cannot repeat again
- c) No, because Ename can be blank
- d) No, because Ename can be duplicate

8. There is an issue near the Position varchar statement. What is it?

- a) It must be int data type not varchar data type
- b) It must be int data type without a proper size
- c) It must be decimal data type
- d) It must have a size if it is varchar data type

9. CHECK constraint used to

- a) Provide a default value for a column
- b) Ensure that all values in a column are different
- c) Uniquely identify each row/record in a database table
- d) Ensure that all the values in a column satisfies certain conditions

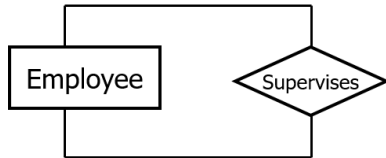
10. Instead of (operation = 'INS' OR operation='DEL') you can use below command

- a) (operation OR ('INS', 'DEL'))
- b) (operation LIKE ('INS' OR 'DEL'))
- c) (operation IN ('INS' OR 'DEL'))
- d) (operation IN ('INS', 'DEL'))

Question Two – Write one word answers only

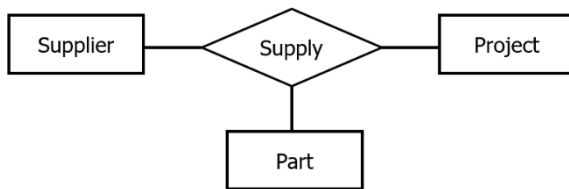
**If you don't write the answers in this paper you won't get any marks for question two.
(Each carry 02 Marks. 2 * 5 = 10 Marks)**

1.



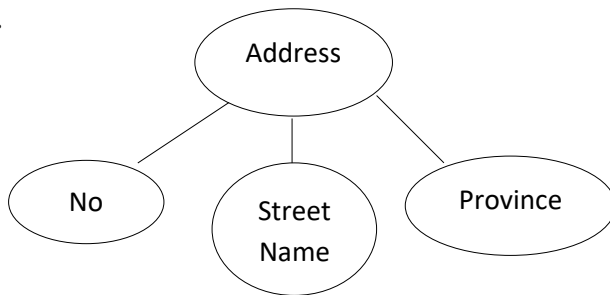
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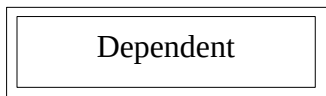
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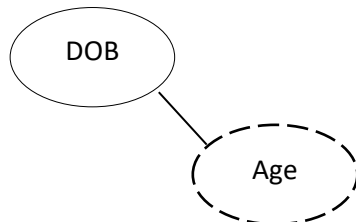
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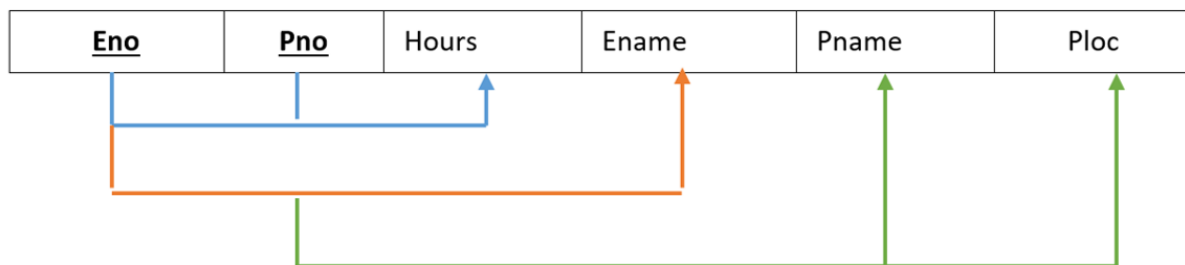
Question Three (10 Marks)

The company is organized into departments. Each department has a name, a unique number, telephone number. Each department is managed by one employee and a department will have only one employee managing it. The DB will keep track of the start date when that employee began managing the department. A department may have several locations. A department controls a number of projects, each of which has a name a unique number and a single location. The DB will store each employees name, unique employee number, address, salary and DOB. An employee is assigned to one department but may work on several projects which are not necessarily controlled by the same department. The DB will keep track of the number of hours per week that an employee work on each project. Each employee is supervised by a supervisor. The DB will keep track of the dependents of each employee for insurance purposes. It will store each dependent's name, DOB, and relationship to the employee.

Draw ER diagram for above requirements. You must mention all your assumptions. (10 marks)

Question Four (20 Marks)

- i) What is Normalization? (04 Marks)
- ii) Describe 1NF, 2NF and 3NF (No need any examples). (06 Marks)
- iii) Consider the below table. Eno and Pno are Primary Keys



- a. Normalize the data shown in this table to Third Normal Form (3NF). (Must show step by step) (10 Marks)

Question Five (20 Marks)

Consider the following Tables. One Software Project has many Business Analysts and One Department has many Business Analysts.

SOFTWARE_PROJECT

<u>Project_Code</u>	Project_Name	Duration
P1	Brandix HR Software	6M
P2	MAS HR Software	4M
P3	CEAT Leaves Software	5M

BUSINESS_ANALYST

<u>Emp_No</u>	Emp_Name	Emp_Experience	Emp_Salary	Proj_ID	Dep_ID
E100	Gayan	02	65000	P1	D1
E200	Waruna	05	98000	P1	D1
E300	Amila	06	125000	P2	D2
E400	Aravinda	01	45000	P3	D2

DEPARTMENT

<u>Dep_No</u>	Dep_Name
D1	Division I
D2	Division II
D3	Division III

Based on the above Tables write SQL Queries for following Requirements. (each carried 2 marks)

1. Create a database called MIT.
2. Create a table called Software_Project with above fields. Give suitable data types.
3. Create a table called Business Analyst with above fields. Give suitable data types. **Do not assign Proj_ID Column as a Foreign Key.**
4. Alter the Business Analyst table and add Proj_ID as a foreign key references to the Software_Project table.
5. Write SQL statement to insert data to Business Analyst table. **(Only one insert query is enough)**
6. Display all Business Analysts details.
7. Display Employee Number and Employee Name of all Business Analysts whose Salary greater than 55000.
8. Display all details of Business Analysts whose Salary between 40000 and 65000 and Experience greater than 03 Years.
9. Display name of Business Analysts whose name start with 'G'.
10. Can you Drop Software_Project Table? Explain.

Question Six (20 Marks)

Based on the above Tables write SQL Queries for following Requirements (use Aggregative Functions, Group by, Having, Joining) to do the following: (Each carried 3 marks)

1. Display the Total Salary of Business Analysts in each Software Project.
2. Display the Average Salary of Business Analysts in each Software Project in the descending order of Average Salary.
3. Display the Software Projects which have Total Salary exceeding 96000.
4. Display Employee Name, Experience and Project Name of all Business Analysts.
5. Display Employee Name, Experience and Project Name of all Business Analysts who get paid between 45000 and 89000.

Database Security (5 marks)

6. Grant All permission for user1 to Business Analyst table (Suppose user1 already created). (2.5 marks)
7. Revoke only Select Permission from user1 from Business Analyst table. (2.5 marks)